

Project Name	Import Data using Transform Maps	Maximum Marks	4 Marks
Date	31 January 2025	Team ID	6ab615fddcdade33acb38af6f

Solution Requirements (Functional & Non-functional)

These requirements describe what the solution must do (functional) and how well it must do it (non-functional) when importing legacy employee training data into ServiceNow with Transform Maps. Each functional requirement is linked to the user stories on the previous design page.

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub-Requirements (Story / Sub-Task)	Linked Stories	Priority
FR-1	Data Source & Staging Load	- Legacy CSV file upload interface. - Scheduled import data source configuration. - Staging table data ingestion (u_training_staging).	USN-7	High
FR-2	Transform Map & Coalesce Matching	- Unique-key coalesce matching (Employee ID / Email). - Field mapping rules for training score and completion date. - Handling of duplicate records and staging error logs.	USN-8, USN-9	High
FR-3	Reference Resolution via Dot-Walking	- Dynamic department and manager reference lookups. - Scripted field mapping for dot-walking relationships. - Validation and error handling for unresolved parent references.	USN-9	High
FR-4	Role-Based Access Control & ACLs	- Custom role creation (x_training_admin, x_training_viewer). - Table-level Read/Write access control lists. - Field-level security for confidential training score results.	USN-10	High

Priorities and story links are suggested by the team and can be adjusted during sprint planning.

Requirement Flow

The four functional requirements follow the order of the import pipeline: **FR-1** loads raw CSV data into staging, **FR-2** matches and maps it to target records, **FR-3** resolves department and manager references, and **FR-4** secures the resulting training records and score fields.

Non-functional Requirements

NFR No.	Category	Description	How It Is Measured
NFR-1	Usability	Intuitive form layouts, clear error messages during Transform Map execution, and guided admin interfaces for data management.	Admin completes an import without external help; every failed row shows a readable reason.
NFR-2	Security	Strict enforcement of role-based access control (RBAC), encrypted data transmission (HTTPS/TLS), and audit logging for record modifications.	Users without the required role cannot read or write protected data; changes appear in the audit log.
NFR-3	Reliability	Transaction rollback on transform script failures, and robust data validation checks before target table ingestion.	A forced script failure leaves the target table unchanged.
NFR-4	Performance	Optimized batch processing for large legacy CSV files (up to 10,000 records), completing transform execution within 3 minutes.	A 10,000-record test file transforms in 3 minutes or less.
NFR-5	Availability	99.9% instance uptime, backed by ServiceNow multi-tenant cloud infrastructure and high-availability server clustering.	Uptime is tracked against the 99.9% target.
NFR-6	Scalability	Ability to scale import processing horizontally through the ServiceNow background job scheduler and asynchronous queues.	Larger or parallel imports run without degrading other instance jobs.

The measures in the last column are suggested acceptance checks for the team to confirm.

Traceability Summary

How the non-functional requirements support each functional requirement.

FR	Requirement	Supporting NFRs
FR-1	Data Source & Staging Load	NFR-1, NFR-4, NFR-6
FR-2	Transform Map & Coalesce Matching	NFR-1, NFR-3, NFR-4
FR-3	Reference Resolution via Dot-Walking	NFR-1, NFR-3
FR-4	Role-Based Access Control & ACLs	NFR-2, NFR-5